

Adyasha Patra

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Education

University of California, San Diego , MS in Computer Science <i>Pursuing Depth Specialization in AI/ML</i>	Graduation: Jan 2027 GPA: 4.0/4.0
Indian Institute of Technology, Bombay, India , BTech in Computer Science <i>Honors in CS and Minor in Artificial Intelligence and Data Science</i>	Graduation: May 2025 GPA: 9.24/10

Publications

Code-Switched Language Identification(LID) for Speech July 2024 – November 2024
Accepted for Publication at EACL 2026 Guide: Prof. Preethi Jyothi

- Developed an innovative solution to address the real-world challenge of routing multilingual voice queries to the appropriate monolingual or code-switched spoken language understanding(SLU) models
- Improved the identification of embedded English in code-switched languages, overcoming limitations of state-of-the-art language identification models in complex multilingual scenarios
- Addressed overfitting of English in monolingual non-English datasets through parameter-efficient finetuning techniques(PEFT) such as LoRA and adapters, utilizing minimal accented English data

Word Grouping for NLP Applications May 2024 – October 2024
[\[Preprint\]](#) Guide: Prof. Ganesh Ramakrishnan

- Proposed a novel word-grouping strategy for Indian languages to unify syntactic structures across parallel sentences, addressing the challenges posed by the agglutinative and inflectional nature of these languages
- Demonstrated the effectiveness of the grouping technique through both intrinsic and extrinsic evaluations, including improved machine translation accuracy from Hindi to five Indian languages
- Surpassed traditional phrasal chunking by offering finer semantic granularity and improved alignment, demonstrated by well-aligned dependency parse trees across multiple languages following word grouping

Research Experience

Improving Performance of Math Prover Agentic Models Oct 2025 - Present
Graduate Researcher | Rose-STL Lab @ UCSD Guide: Prof. Rose Yu

- Surveyed state-of-the-art Lean4 theorem provers & datasets (DeepSeek-Prover, Gödel, Kimina, NuminaMath)
- Enhancing informal mathematical reasoning in theorem-proving models by transforming Lean4 formal proofs into informal derivations and training Process Reward Models (PRMs) for step-level reasoning evaluation

Faster LLM Reasoning Using Speculative Tool Invocation Sept 2025 - Dec 2025
Graduate Researcher | WukLab @ UCSD Guide: Prof. Yiyang Zhang

- Designed & benchmarked a dual-model speculative tool-calling framework (Speculator-Actor) on the GAIA dataset, for pre-execution & caching of safe read-only tools to accelerate LLM agent reasoning
- Performed knowledge distillation on Qwen3-4B using tool-calling trajectories generated by Gemini-2.5-Pro

ASR to Support Users for Speech Pathologies Jan 2025 – May 2025
Undergraduate Researcher | CSALT @ IIT Bombay Guide: Prof. Preethi Jyothi

- Motivated by Google's Project Euphonia, approached the challenge of personalized ASR for pathological speech by proposing Single Utterance Test-time Adaptation (SUTA) on Wav2Vec 2.0, achieving substantial improvements on the Torgo dataset for dysarthric speakers
- Independently designed a SUTA pipeline that adapts the feature extractor and layer norm layers during inference, and benchmarked it against LoRA-based fine-tuning of Whisper for dysarthric speech

Multilingual Pre-tokenizer Jan 2024 – June 2024
Undergraduate Researcher | BharatGen @ IIT Bombay Guide: Prof. Ganesh Ramakrishnan

- Enhanced multilingual tokenization for Indian languages by developing a pre-tokenization strategy to address challenges such as suffix-prefix variations and sandhi, key features of Indian linguistic structures

- Combined the pre-tokenization pipeline with Byte Pair Encoding (BPE) and evaluated the impact of the strategy by analyzing enhancements in fertility scores & continuation rate
- Validated on the IndicCORP-v2 dataset, containing 9.89 million unique Hindi words with expert linguist review

Cross-lingual Transfer Learning

Jan 2024 – June 2024

Undergraduate Researcher | CSALT @ IIT Bombay

Guide: Prof. Preethi Jyothi

- Performed an extensive literature review of state-of-the-art multilingual models and downstream NLP tasks
- Analyzed zero-shot cross-lingual transfer performance by fine-tuning multilingual XLM-R and mT5 models across 11 languages, demonstrating improved transfer when source and target languages share linguistic similarities
- Developed an instance-level finetuning approach using expert models, enhancing accuracy in the target language

Internships

Technical Intern (CTO's Office)

July 2025 – Sept 2025

Prodigal (YC W18)

Mumbai, India

- Developed a multi-agent autonomous system that takes inputs from Google Sheets, Databricks, & GitHub, while parsing Asana chats for context, to automatically generate dbt models & transformation workflows in SQL
- Automated dbt lifecycle, PRs, & GitHub commits, enabling iterative feedback loops with minimal manual effort
- Optimized production workflow, reducing run-time by 40% via Databricks monitoring and fleet cluster migration

Research Intern

May 2025 - June 2025

Zingle AI

Remote

- Conducted comparative analysis of code agents (Copilot, Cursor, OpenAI Codex, Claude Code) for database documentation in large codebases, assessing correctness, coverage, and usability to guide tooling adoption
- Studied core design principles and workflows of Windsurf and Cursor IDEs to evaluate the need for developing a new IDE for the startup, supporting strategic decision-making
- Analyzed agentic tool call patterns and inefficiencies to better understand existing workflows and identify gaps in current IDE solutions

Summer Analyst | Software Developer

May 2024 - July 2024

Goldman Sachs

Bangalore, India

- Engineered a robust and scalable approval workflow mechanism for the firm's inventory management platform, using Java and Spring Boot to streamline operations and ensure seamless adaptability
- Enhanced risk mitigation by implementing an additional level of check, reducing the potential for critical errors
- Accelerated feature onboarding by developing an extendible & modular framework, reducing development time

Student Researcher | Guide: Prof. Andreas Waag

May 2023 – July 2023

Technical University of Braunschweig

Braunschweig, Germany

- Enhanced lensless microscope magnification to 1.26 by multi-tilt superimposition & Angular Spectrum Method
- Implemented a system to detect rotation angles using the SIFT algorithm, achieving 94.67% accuracy for USAF
- Developed a robust GUI for holographic microscopy, supporting multi-object segmentation & real-time analysis

Technical Projects

Dynamic Agentic RAG | InterIIT Tech | Problem Statement Lead

October 2024 - December 2024

InterIIT Gold for Problem Statement

IIT Bombay

- Designed a dynamic Retrieval-Augmented Generation (RAG) system for efficient query resolution and analysis, simultaneously accounting for token efficiency tailored to the financial domain
- Implemented a novel decomposer framework, build upwards from Contragen to iteratively break down multi-hop queries into parallelizable units, leveraging a critic-generator model for optimal performance
- Explored persona-based multi-agent collaboration for complex analytical queries, incorporating supervisor-agent and swarm interaction paradigms utilizing LangGraph for architecture and LangSmith for testing purposes
- Built a LLM model agnostic and embedding model agnostic architecture, with rigorous testing over 5 open-source and closed-source LLMs and 3 finance-specific and general embedding models

Incorporating Rewrite Feedback in RLHF | Human Centered AI | IIT Bombay Jan 2025 - May 2025

- Designed and implemented a synthetic dataset generation pipeline leveraging large language models (Llama-2 and Gemini) to produce paired responses and rewrite-based feedback for reward model training in RLHF
- Developed a binary and rewrite-based scalar reward model using pairwise preference loss to effectively distinguish and rank responses, improving evaluation and fine-tuning of language models
- Automated response evaluation and rewriting by integrating LLM-based preference judgments and rewrite generation, simulating high-quality human feedback under limited data constraints

Contrastive and Graph-based Learning

Organization of Web Information | IIT Bombay Jan 2025 - May 2025

- Fine-tuned stance-aware sentence transformers using Siamese and Triplet contrastive learning to distinguish opposing viewpoints in Kialo debates
- Explored information bottleneck trade-offs by compressing BERT representations for intent classification on CLINC-150, incorporating autoencoders and stochastic noise to analyze performance vs. compression
- Enhanced multi-hop QA retrieval by constructing a Graph of Passages and leveraging Graph Neural Networks to model inter-passage relationships, improving passage ranking over standard dense and sparse retrieval methods

ML for Retrieval | Information Retrieval | IIT Bombay August 2024 - November 2024

- Optimized large-scale image retrieval using neural locality-sensitive hashing (LSH), leveraging multiple hash tables to mitigate collisions and enhance scalability for complex datasets
- Designed & trained GNN models for large-scale network analysis & link prediction, integrating LSH for inference
- Improved retrieval performance on HotpotQA and WikiNQ datasets by fine-tuning BERT models with query and document likelihood approaches, boosting Mean Average Precision (MAP) by 20% compared to baseline

Pre-emptive Shielding in RL with Blackout Constraints

Formal Methods for ML | IIT Bombay August 2024 - November 2024

- Designed and implemented a pre-emptive delta-shield mechanism for the Pacman agent to recommend safe actions under partial observability, including blackout intervals limiting adversary state visibility
- Extended and modified an existing probabilistic shield codebase by integrating blackout scenarios, adapting state-space representation and shield logic to handle uncertainty during blackouts

Shape Prior Segmentation | Medical Image Computing | IIT Bombay January 2024 - May 2024

- Improved segmentation on ACDC heart dataset using Shape Prior Module, combining local & global shape priors
- Enhanced scalability, ensuring robust performance of SPM across CNNs & Transformer-based architectures

Compiler Design | Implementation of Programming Languages | IIT Bombay January 2024 - May 2024

- Built a comprehensive compiler for C-like language using Lex and Yacc, ensuring accurate syntax analysis
- Designed and implemented key compiler components, such as lexical analysis, parsing, abstract syntax trees, intermediate code, and register transfer language, culminating in the generation of MIPS Assembly code

Text Style Transfer | AI and ML | IIT Bombay August 2023 - November 2023

- Investigated Transformer-based architectures for text style transfer without explicit latent space disentanglement
- Conducted reproducibility with architecture modifications, analyzing the impact of changes on text attributes

Coherence-Enhancing Diffusion | Digital Image Processing | IIT Bombay August 2023 - November 2023

- Implemented Coherence-Enhancing Diffusion for better flow and completion of interrupted lines in images
- Extended the algorithm to color spaces and conducted a comparative analysis of HSI and RGB models

Optimal Cue-Stick Control | Reinforcement Learning | IIT Bombay August 2023 - November 2023

- Developed a reinforcement learning-based billiards agent using Python to optimize cue-stick control
- Utilized state estimation and predictive modeling to improve the agent's accuracy across various game levels

Minimax Graph Solver | Design & Analysis of Algorithms | IIT Bombay January 2023 - May 2023

- Engineered a minimax dynamic programming algorithm for a two-player weighted directed graph game
- Implemented memoization to determine optimal strategies, achieving efficient space & time complexity

Railway Planner | Data Structures and Algorithms Lab | IIT Bombay August 2022 - November 2022

- Developed an interactive Railway Planner in C++ utilizing various data structures namely Trees, Graphs, Priority queues, Dictionaries and Tries and algorithms, such as QuickSort and KMP Pattern matching
- Utilized Multigraph model and customized Dijkstra's algorithm for optimal route finding with constraints

Cinemas A-Z | Software Systems Lab | IIT Bombay August 2022 - November 2022

- Built a web app to view movie and TV show details from IMDb, Rotten Tomatoes, and Metacritic in one platform
- Added 1,000+ movies via web scraping with BeautifulSoup and included a live scraper for real-time updates
- Developed a user database for account management, allowing dynamic recommendations based on user activity

Scholastic Achievements

- Honoured with the Desai Sethi Scholarship by IIT Bombay in recognition of academic achievements (2023)
- Secured All India Rank of 68 in Joint Entrance Examination JEE Main among 1 million students (2021)
- Achieved All India Rank 284 in JEE Advanced among 0.15 million candidates (2021)
- Awarded the Kishore Vaigyanic Protsahan Yojana KVPY fellowship with an All India Rank of 67 (2020)

Positions of Responsibility

General Secretary May 2024 – May 2025

Computer Science and Engineering Association, IIT Bombay

- Spearheading a council of 20 members in organizing department cultural and sports events throughout the year
- Managing a budget of 1 million INR, ensuring effective allocation of resources and successful event execution

Institute Student Mentor and Department Academic Mentor May 2023 – May 2025

Institute Student Mentor Program, IIT Bombay

- Mentoring 12 freshmen by providing guidance and support to help them acclimate to college life
- Mentored 6 sophomores providing them necessary help and guidance regarding academics and internships

Teaching Assistant June 2023 – June 2024

CS663: Foundations of Digital Image Processing, Fall 2024, IIT Bombay

- Assisted in providing academic support and graded assignments for around 230 students

Class Representative June 2023 – June 2024

Computer Science & Engineering Department, IIT Bombay

- Serving as the primary point of contact between professors, CSE council and a batch of 180+ undergraduates
- Putting forward the concerns of the students to authorities and helping professors in smooth conduct of courses

Technical Skills

Programming	C/C++, Python, Java, Bash
Data Science	Numpy, Pandas, Matplotlib, PyTorch, MATLAB, OpenCV
Development	Spring Boot, Kafka, HTML, CSS, JavaScript, Git, TeX

Languages Known

English (**Proficient**) Oriya (**Native**) Hindi (**Fluent**) Bengali (**Fluent**)

Extracurricular Activities

- Selected among the top 35 teams from 500 for the first-ever Peak XV Summit Hack: Consumer AI (2025)
- Attended the two-day Google DeepMind Research Symposium in Bangalore, India (2025)
- Volunteered in community service and built solar lamps for underprivileged areas (2024)
- Travelled to 11 European countries experiencing diverse cultures with a group of fellow students (2023)
- Completed a year-long training course in Badminton through National Sports Organization NSO (2022)